CityNexus - Smart City Traveler System | System Architecture

CNX-SE-VIS-01 System architecture (layered)
Rev A - generated 2026-10-22 - source tree CityNexus @ review/boli-party-occupancy-scope-r1, github.com/Canshu-King/CityNexus

How the product actually runs. A traveler request enters the React shell, is served by one FastAPI monolith, and may be reasoned about by an AI agent that owns no domain truth: every agent mutation passes an allowlist, an audit write and an authorization decision before a deterministic solver touches journey state.

1 USERS & CLIENT SURFACE

Traveler
phone, tablet landscape, desktop browser
frontend/src/main.tsx:38

Administrator
platform and provider console
use-cases/use-cases.yaml:17

HTTPS + PWA

2 FRONTEND React 19 + TypeScript 5.8 + Vite 7 + PWA (frontend/src)

CopilotDock + AgentComposer
the natural-language input box
features/agent/CopilotDock.tsx:1

AgentRunRail + feed
shows the agent working state
features/agent/AgentRunRail.tsx:1

V8Experience shell
offline snapshot, execution clock, wallet
app/V8Experience.tsx:1

Plan comparison + Trip Studio
accept or reject what the agent proposes
features/planning/PlanComparison.tsx:1

JourneyMap
MapLibre 5.24, OpenFreeMap style, provenance badges
components/JourneyMap.tsx:1

15 routed panels
hub, live, map, studio, stay, discover, agent, twin, footprints, access
routes/v8Routes.ts:1-33

POST /copilot/messages
journey CRUD
console panels
accept / reject
route evidence
experience + twin

3 BACKEND HTTP SURFACE one FastAPI monolith, 175 endpoints under /api/v1 (backend/app/main.py, 14,569 lines)

Agent surface
/copilot/messages, /agent/commands, /agent/tools/{name}, /approvals/{id}/approve, /agent-feed; the MCP V1 and V2 surfaces reach the same gateway
main.py:14169, 1367, 1246, 2081, 10125

Shared transport
journeys, plans, places, replans, check-ins, collections, share, bookings - 175 endpoints
main.py:3136, 7369, 7328, 10923, 4296, 14404

Data & ops surface
/experience, /twin, /route-evidence, SSE /operations/stream, OpenAPI, server-owned agent identity
main.py:5473, 9846, 10525, 1982 ; contracts/openapi.yaml

4a AI AGENT & HERMES PLANE the agent owns no domain truth (backend/app/agent)

Entry: intent + context
8 intents over a server-built context; semantic envelope decides what the model may conclude
main.py:14190-14228, 13593 ; semantic_compiler.py:123,229

Mission, plan, tool loop
compile_mission, build_mission_plan, run(max_steps=12)
agent/mission.py:231,384 ; runtime.py:62,336

Tool gateway
allowlist of 11 tools, audit write, idempotency, then ALLOW_AUTO / REQUIRE_CONFIRMATION / DENY
agent/gateway.py:357-398, 59, 71, 338-347

model turn
tool call
request_replan / accept_replan

4b DOMAIN SERVICES deterministic solver, business facts, journey state machine

Journeys, plans, replans
lifecycle, deterministic seeded solver, proposal-only replanning, accept / reject
main.py:3136, 7369, 7790, 10923, 13097 ; app/replanning

Twin, capability, preference
three preference scopes; declared-not-invented budget and party size
main.py:9846, 2761, 2568 ; models.py:95,121,824

Places, city services, commerce, guardian
real-city discovery, catalog hydration, bookings and commitments, autopilot
main.py:7328, 5558, 5572, 14404, 1599 ; discovery/real_city.py

ORM
twin + versions

5a PROVIDERS & INTEGRATIONS 9 clients; the deterministic mocks are the shipped default

DeepSeek Responses API
chat, vision, web search; ai_enabled defaults to false
integrations/ai/responses.py:244 ; config.py:14-30

Shipped default provider set
mock city / mobility / booking / payment; real-city mode=replay
.env.example:18-31 ; config.py:79-81 ; integrations/mock/providers.py:13

Photon, Nominatim, Overpass
geocode, POI, weather; Valhalla live URL not set
photon.py:20 ; real_city.py:617 ; BLOCKERS.md:13

5b PERSISTENCE, AUDIT & OPERATIONAL STATE SQLAlchemy + SQLite (WAL), 60 tables

SQLite citynexus.db
WAL, foreign keys ON, write deadline and busy timeout
db/session.py:70-88 ; config.py:10, 96-98

Journey truth
journeys, plan_versions, activities, commitments
models.py:175, 511, 535, 578, 712

Agent state and audit
agent_runs, agent_run_steps, approvals, audit_logs
models.py:254, 271, 562, 677

Legend

- Implemented + evidence (test / published commit)
- Implemented, not yet proven on the line
- Documented intent - no implementation here
- External provider / third-party boundary
- Persistence, audit and operational state
- Human actor / client surface
- Agent control plane

THE 11 TOOLS THE GATEWAY ACTUALLY EXPOSES (backend/app/agent/tool_catalog.py:26-59)
read-only: inspect_journey, inspect_twin, list_replans, search_city_services
plan-shaping: request_replan, accept_replan
typed actions: replace_activity, reschedule_activity, create_booking, cancel_refundable_booking, request_commitment_change

NOT AN AUTOMATION CLOSURE: native Hermes binary absent and runtime identity still DIRECT (BLOCKERS.md:9-12); live Valhalla URL not configured (BLOCKERS.md:13); no real payment instrument registered (AGENT_TOOL_CATALOG.md:16-18); impact resolver default disabled (config.py:33); physical phone / PWA shadow trial not performed (CURRENT_STATUS.md:4).

COUNTS, taken from this tree on 2026-10-22: 175 = @app.get|post|patch|put|delete decorators in backend/app/main.py; 60 = __tablename__ declarations in backend/app/db/models.py; 14,569 = lines of main.py; 242 backend test modules, 184 frontend e2e specs.

DEFAULTS: CITYNEXUS_AI_ENABLED=false, CITYNEXUS_REAL_CITY_ENABLED=false, real-city mode=replay, impact provider=disabled (core/config.py:14, 79-81, 33) - the picture describes wiring present in code, not a claim that every provider is live in a given deployment.

AUDIT: every shape carries its code anchor as a small second line and stores the same anchor in the custom property SourceRef, so the drawing can be checked shape by shape inside Visio.